

PROJECT INTRODUCTION:

This project focuses on analyzing an E-Commerce business using Power BI. The company sells products across multiple categories and serves customers from different cities across India through various payment methods.

PROJECT OBJECTIVE:

The objective is to analyze sales performance, customer behaviour, product and category performance, order and payment patterns, customer retention, and revenue contribution.

KEY BUSINESS AREAS:

- Sales and order trends
- Customer purchasing behaviour
- Product and category performance
- Payment and order-status patterns
- Customer retention and repeat purchasing
- Revenue contribution and business growth opportunities

BI ANALYST ROLE:

As a BI Analyst, the objective is to transform raw transactional data into a complete Business Intelligence solution through data cleaning, data modelling, KPI development, DAX analysis, dashboard creation, business insights and actionable recommendations.

PROJECT FLOW:

Understand → Explore → Clean → Model → Measure → Analyze → Visualize → Interpret → Recommend

Analyze Sales Performance

Total Revenue

\$116,154K

Total Orders

1K

Average Order Value

\$232.3083716K

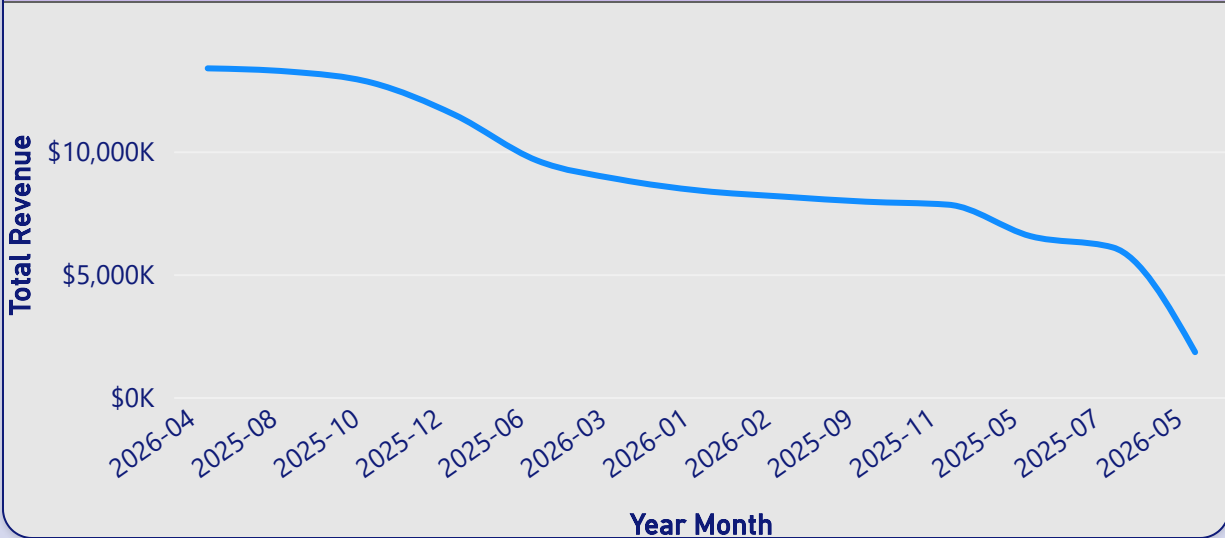
Total Quantity Sold

5K

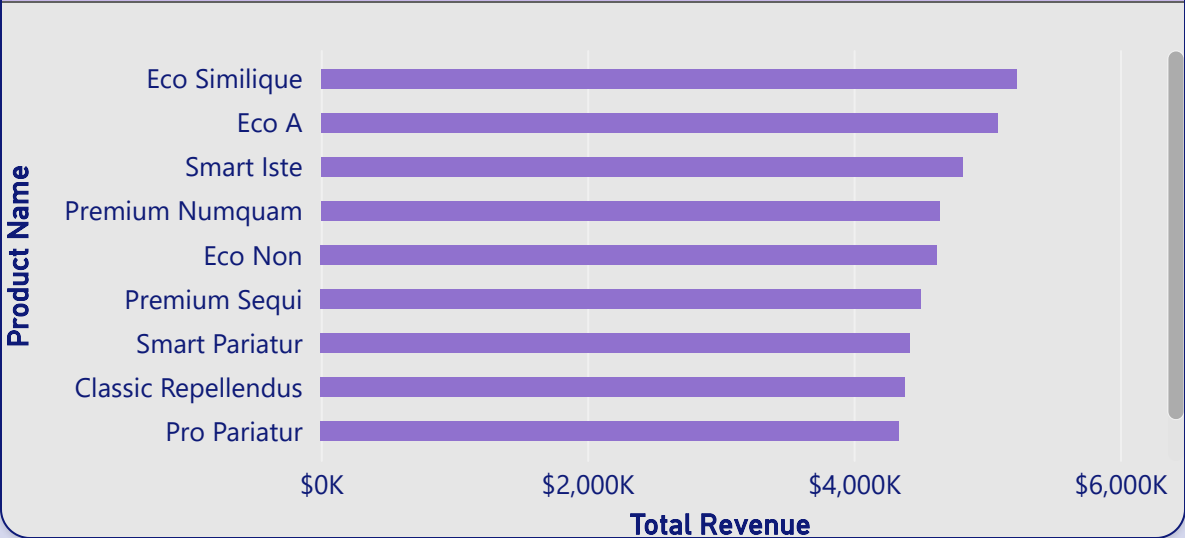
Revenue Growth %

1.60%

Total Revenue by Year Month



Total Revenue by Product Name



Date

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Category

Books

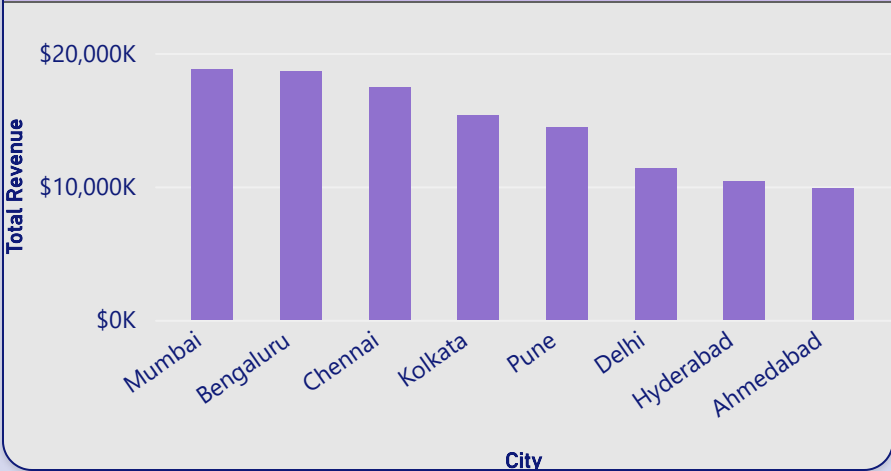
Home & Kitchen

Clothing

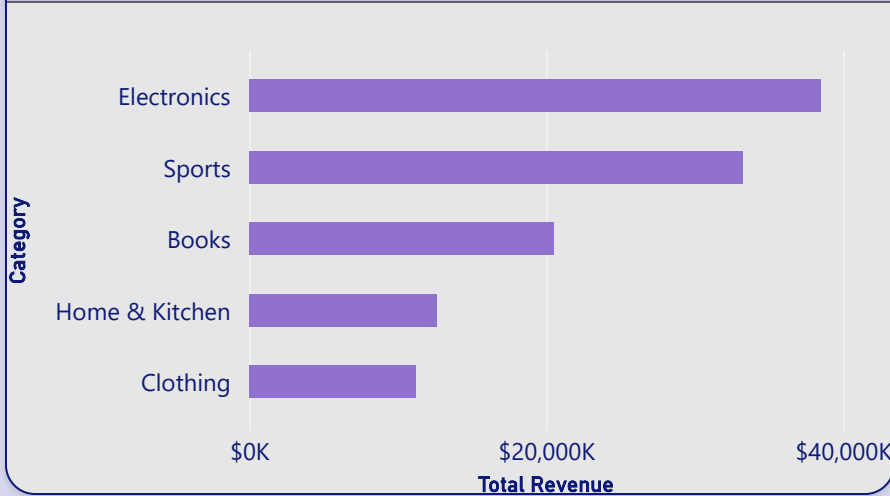
Sports

Electronics

Total Revenue by City



Total Revenue by Category



City

Ahmedabad

Delhi

Mumbai

Bengaluru

Hyderabad

Pune

Chennai

Kolkata

Order Status

☐ Cancelled
☐ Delivered
☐ Pending
☐ Returned

Sales Performance Dashboard — Business Insights

- Total revenue reached approximately **\$116.15M**.
- **Electronics** is the top revenue-generating category.
- Revenue growth stands at **1.60%**, indicating modest positive growth.
- Revenue shows a **declining trend in the later months**.
- City-wise performance varies, highlighting opportunities to improve weaker markets.

Customer Behaviour

Total Customers

100

Customers with orders

99

Repeat Customers

95

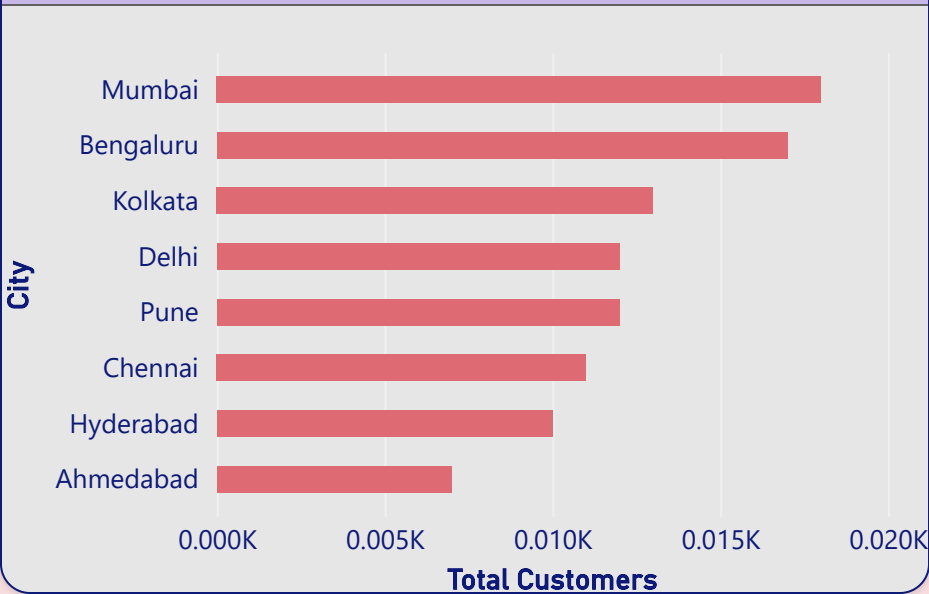
Repeat Customer Rate %

95.96%

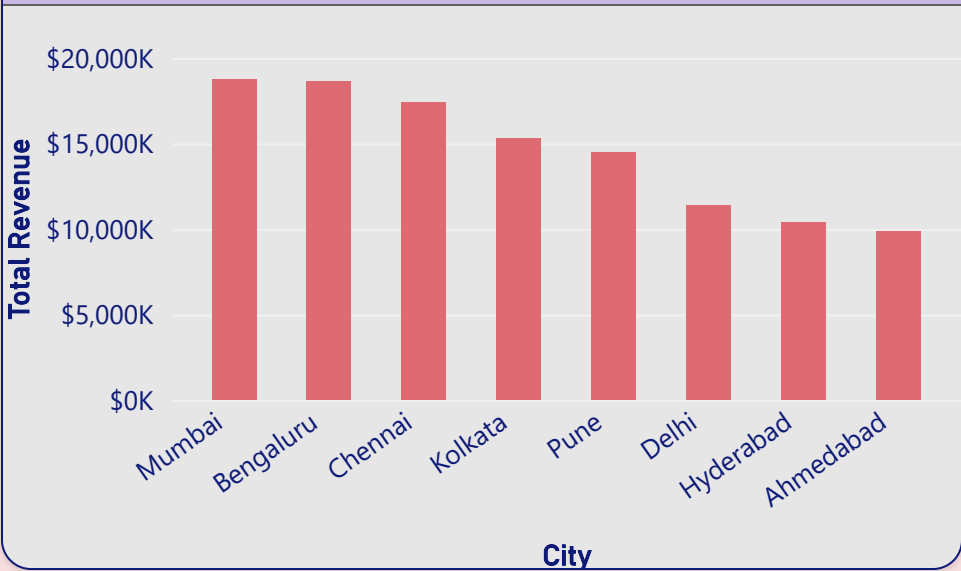
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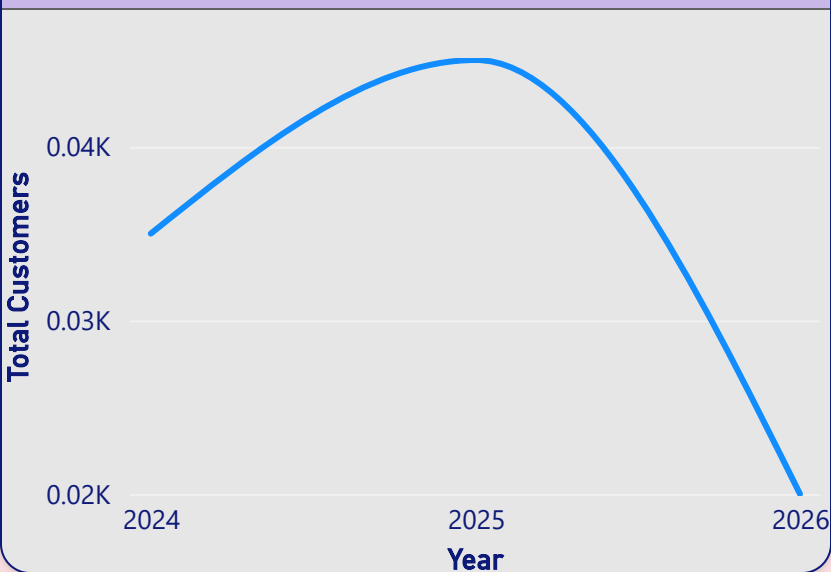
Total Customers by City



Total Revenue by City



Customer Signup Trend



City

Ahmedabad

Delhi

Bengaluru

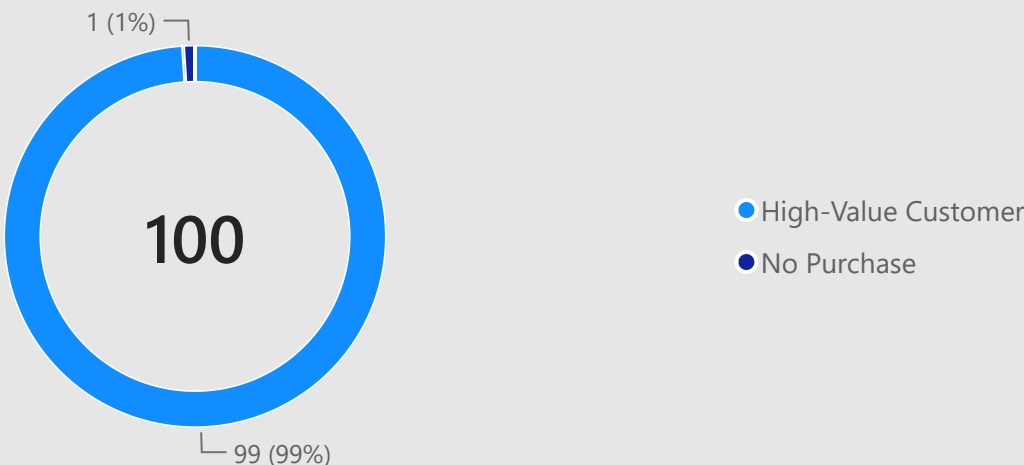
Hyderabad

Chennai

Kolkata

Customer_Name	City	Total Orders	Total Revenue
Balveer Bumb	Chennai	9	\$3,120,955
Harrison Rai	Bengaluru	11	\$2,578,163
Jeet Radhakrishnan	Kolkata	8	\$2,565,21
Thomas Amble	Bengaluru	8	\$2,285,401
Udant Bajwa	Kolkata	9	\$2,220,904
Arunima Dugal	Chennai	7	\$2,216,95
Total		500	\$116,154,185.

Customer Segments



Customer Segment

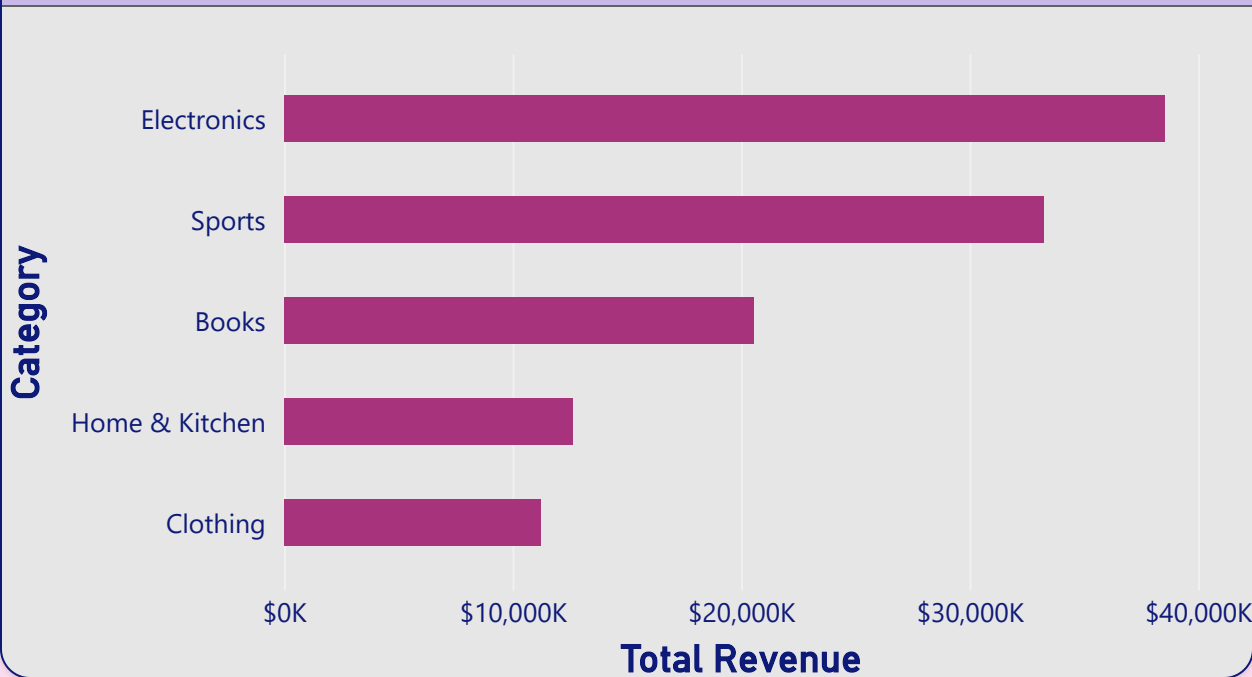
- ☐ Loyal
- ☐ One-Time
- ☐ Repeat

Customer Behaviour – Business Insights

- **100 customers** are registered, with **99 customers placing orders**.
- **95 customers are repeat purchasers**, giving a **95.96% repeat customer rate**.
- **Mumbai has the highest customer base** among the displayed cities.
- Customer signup activity **peaked and then declined**, indicating a need to strengthen new customer acquisition.
- Revenue contribution varies by city, with some cities showing stronger customer value than others.

Product & Category Performance

Total Revenue by Category



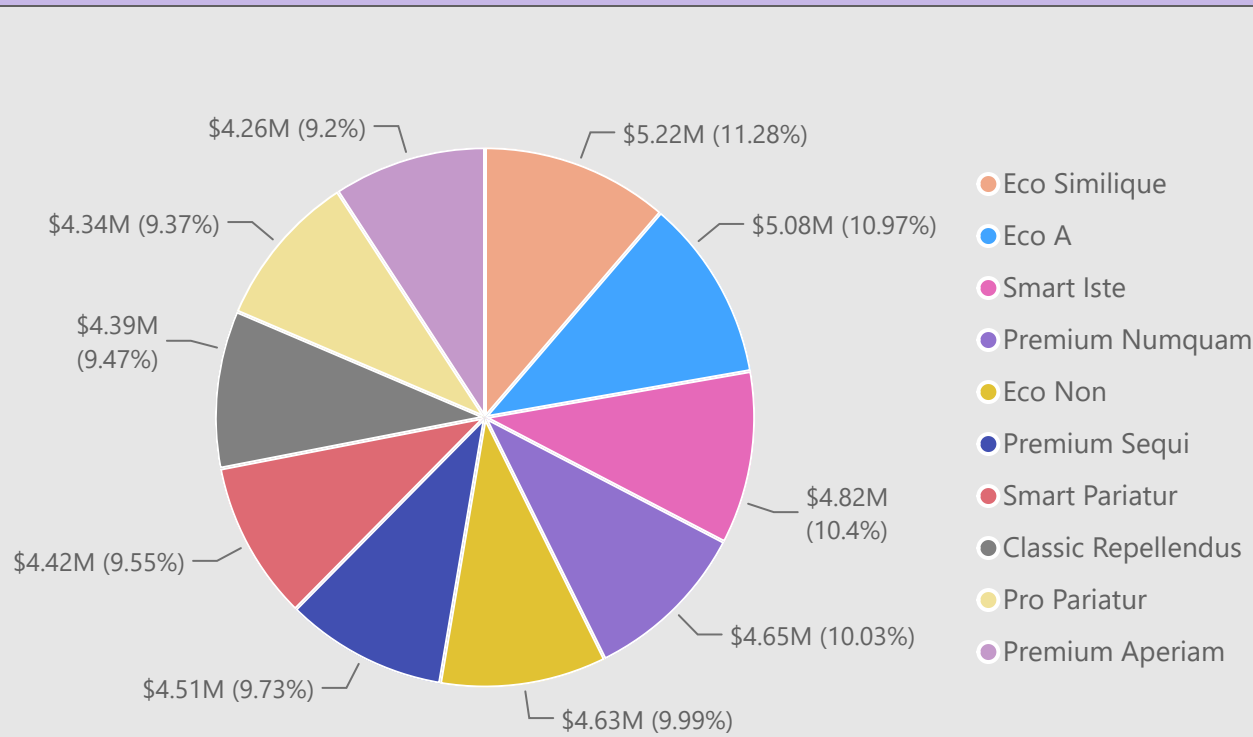
Total Quantity by Product Name



Low Stock Products



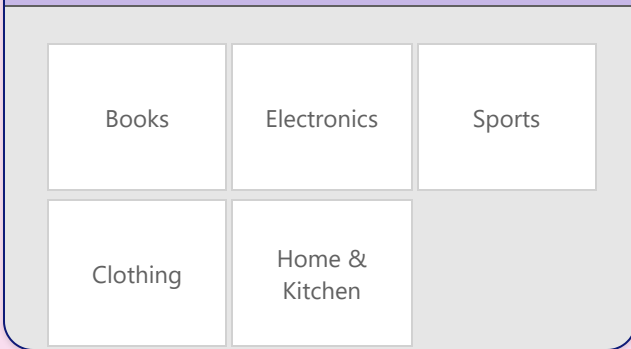
Top Products by Revenue



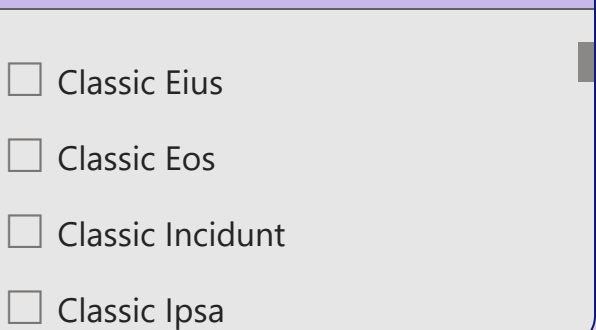
Low-Selling Products



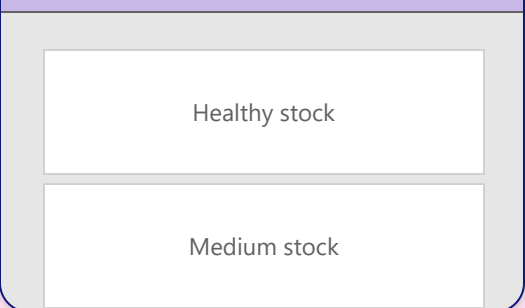
Category

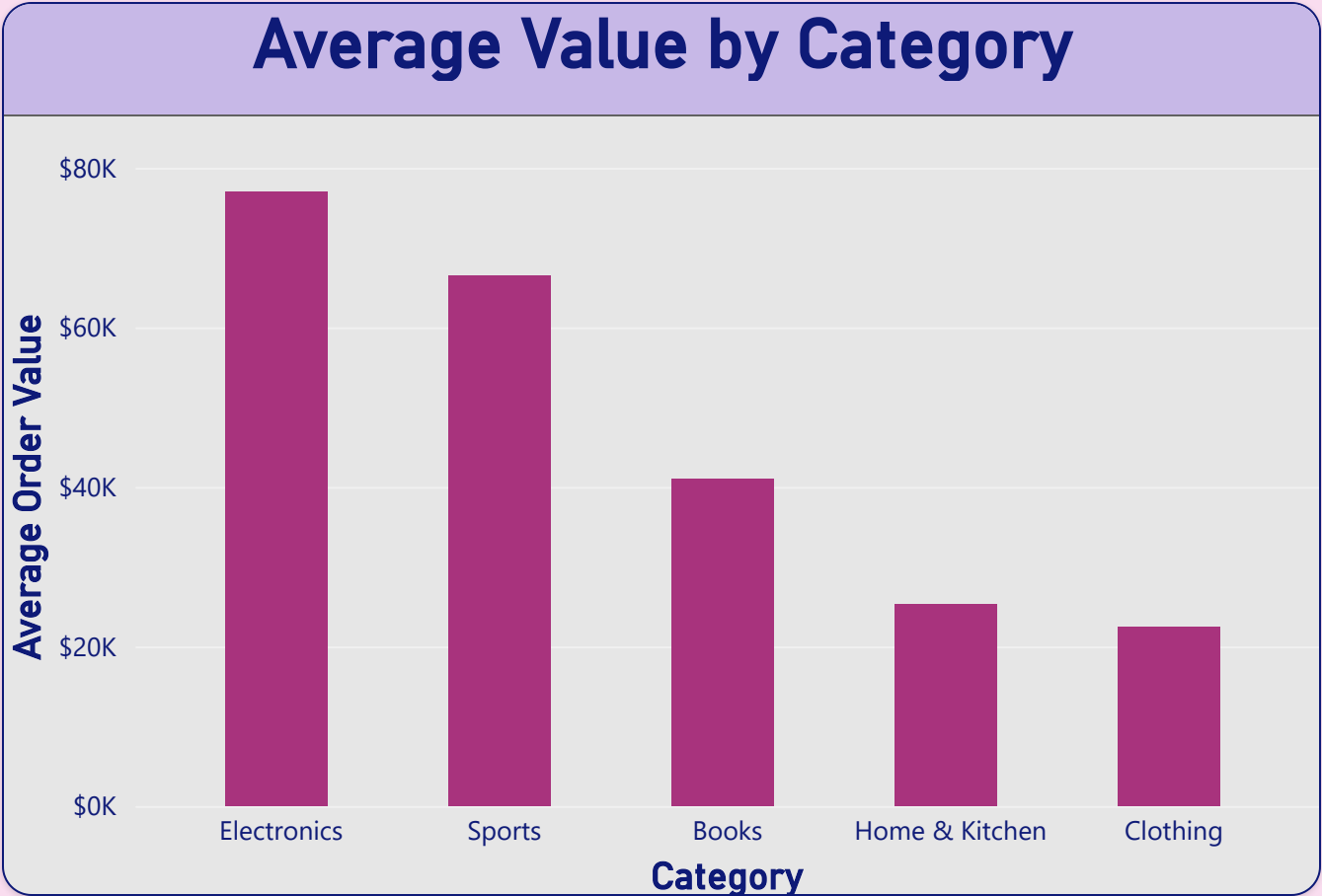
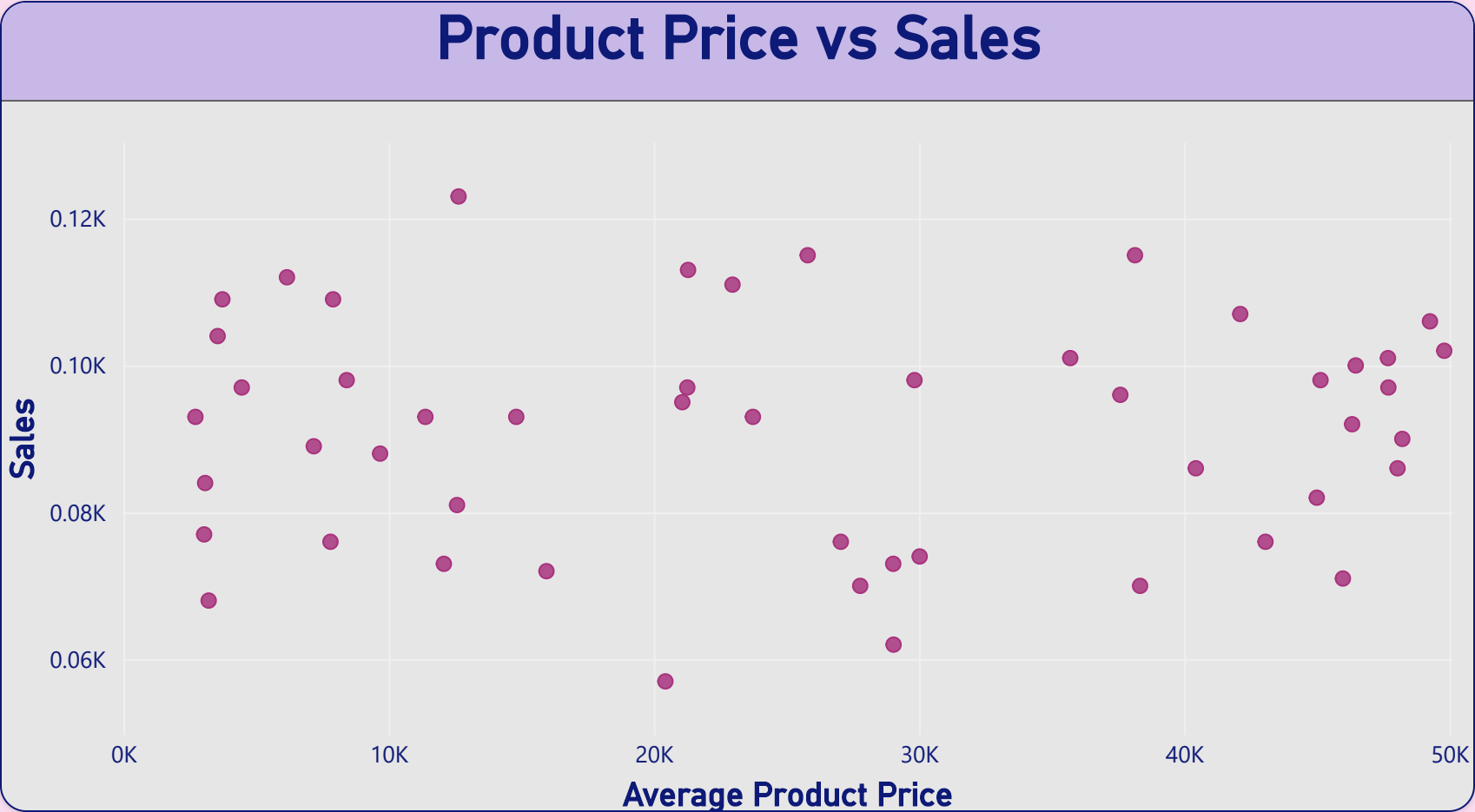


Product Name



Stock Status



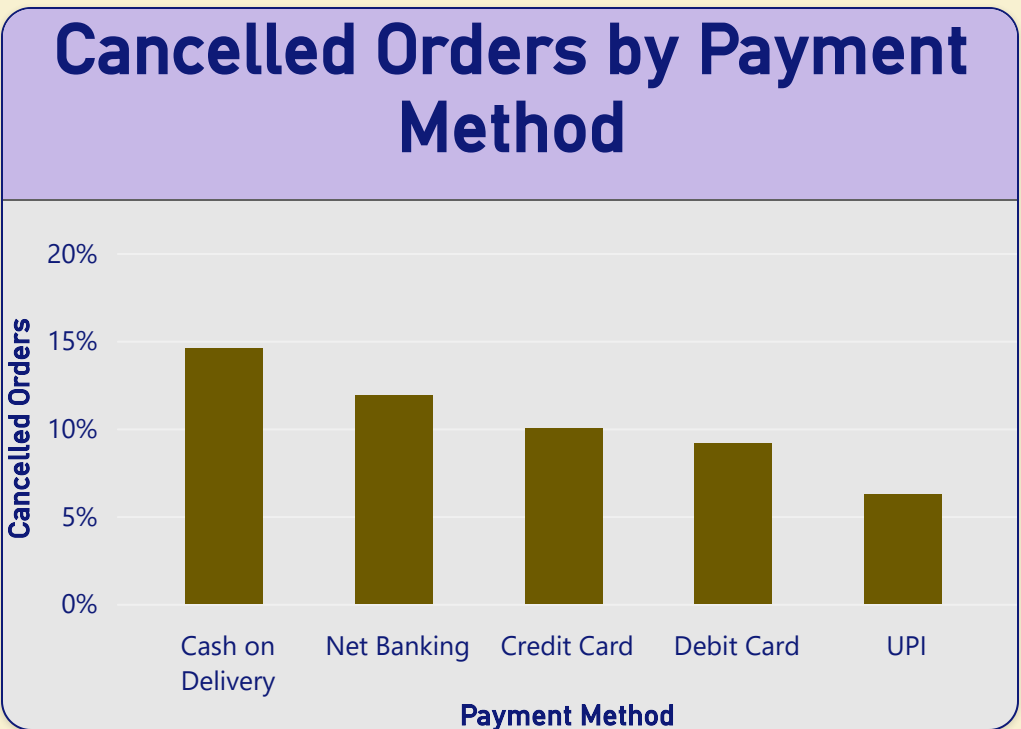
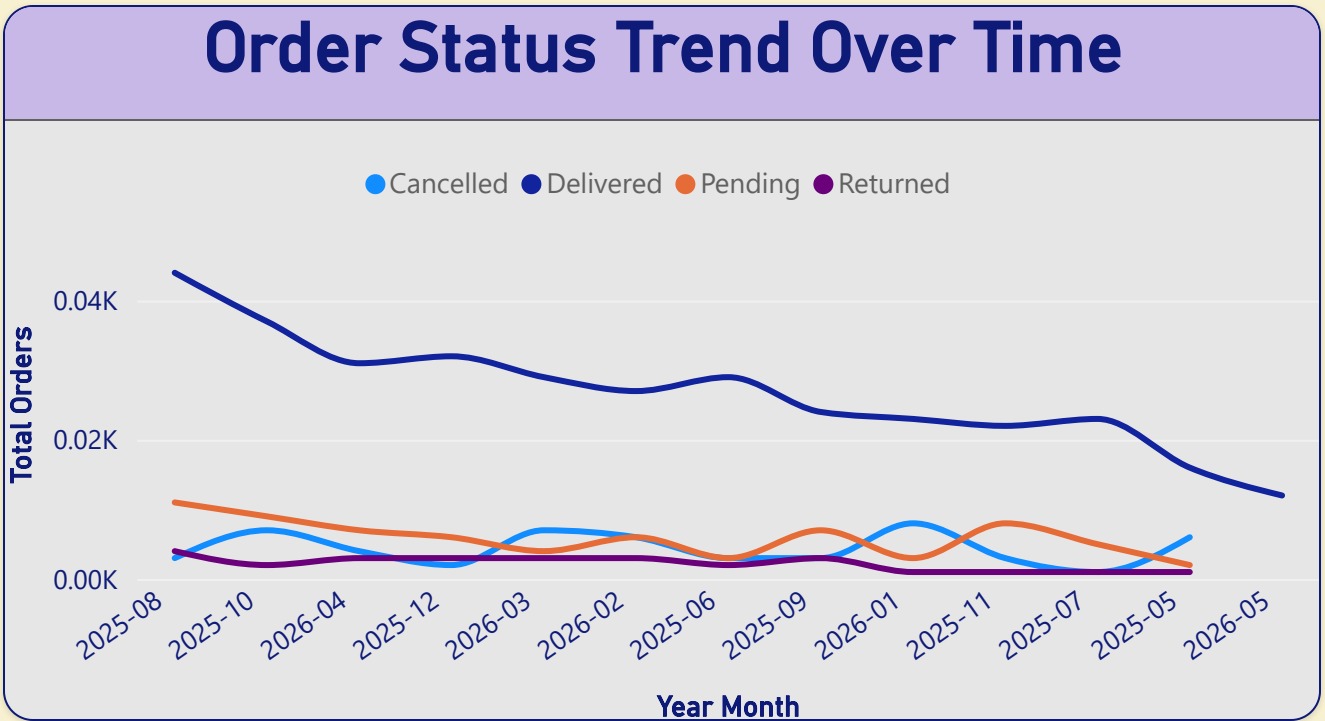
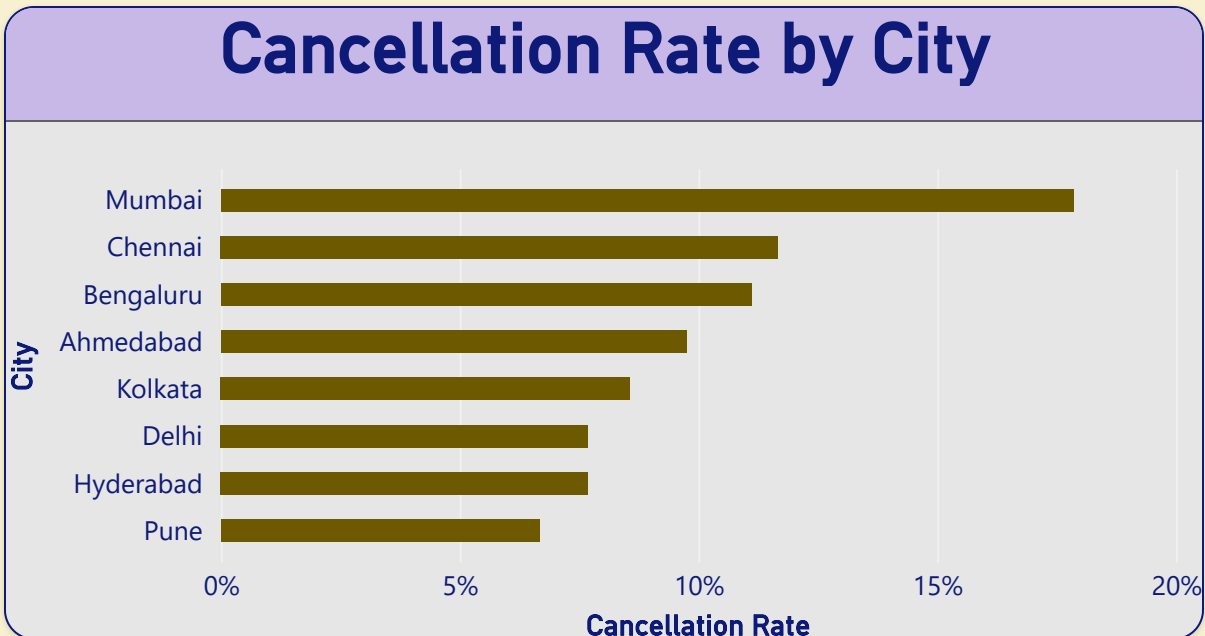
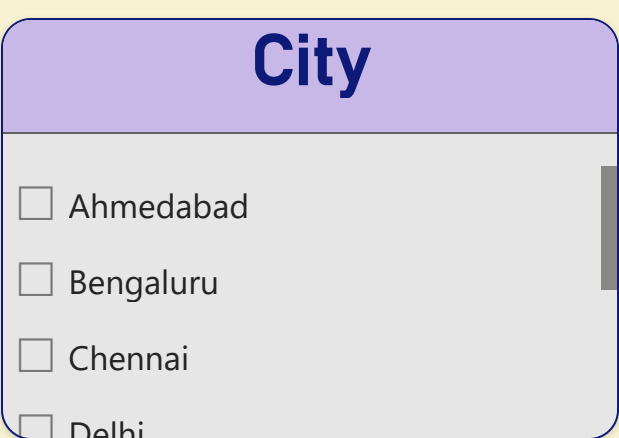
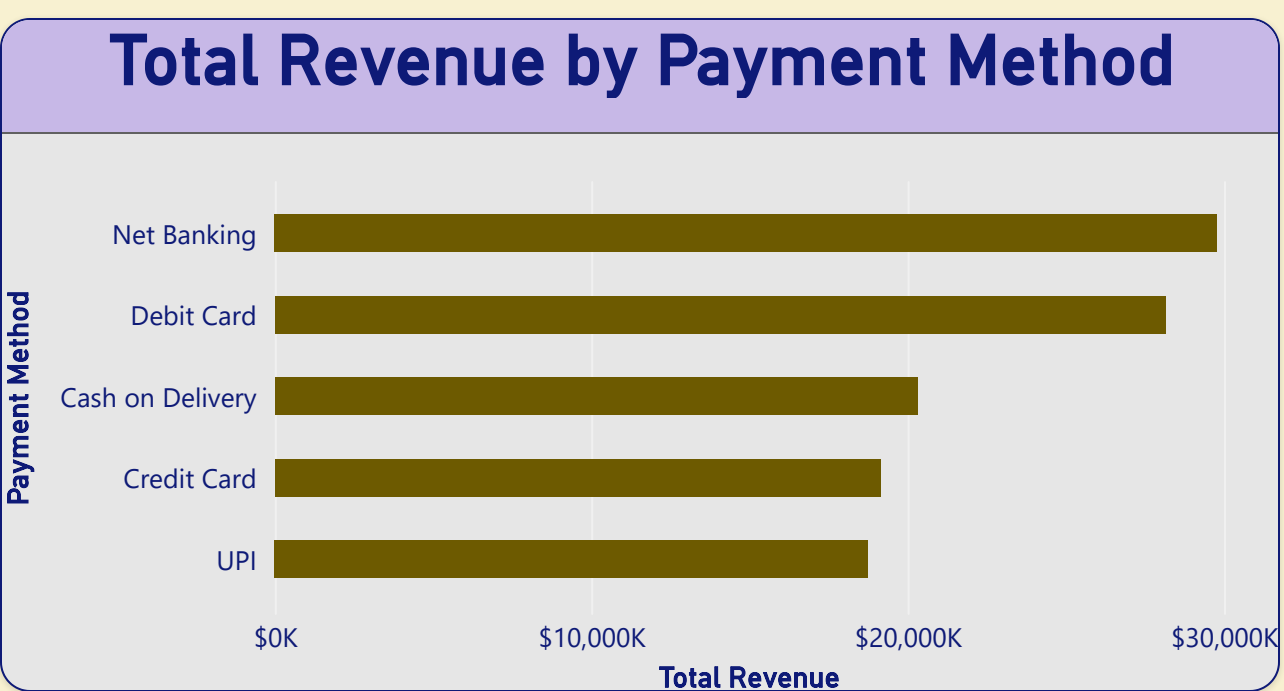
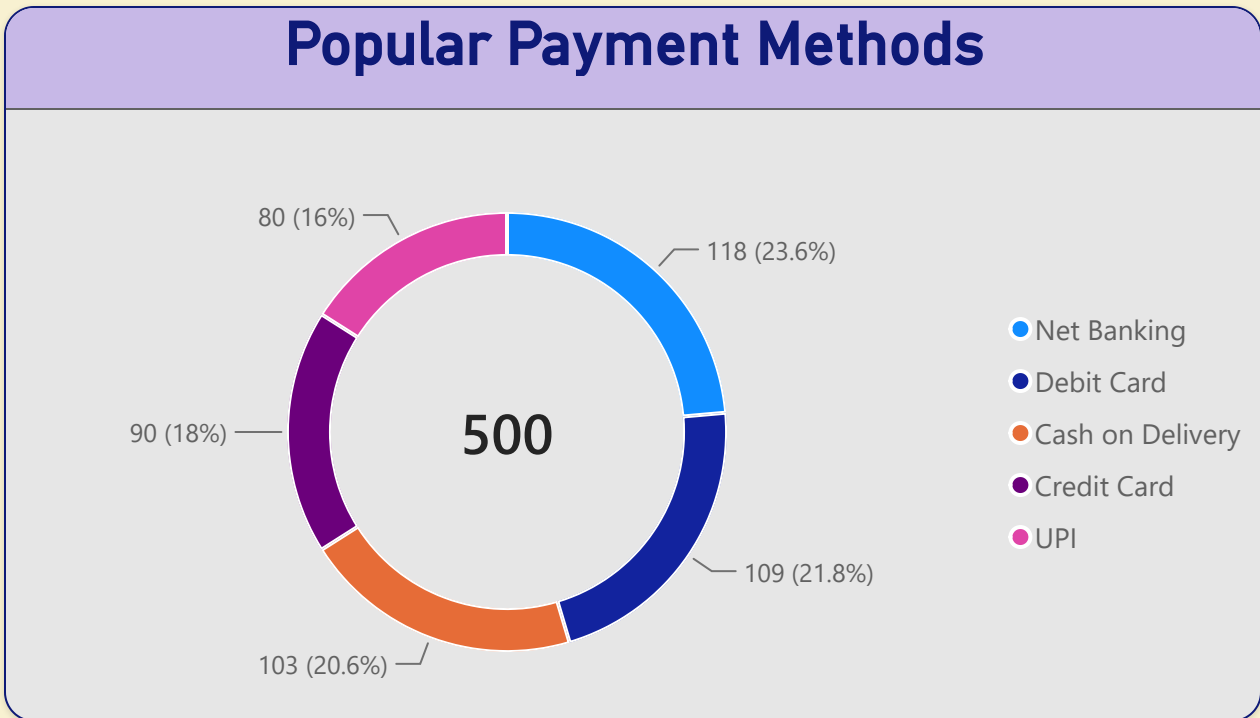
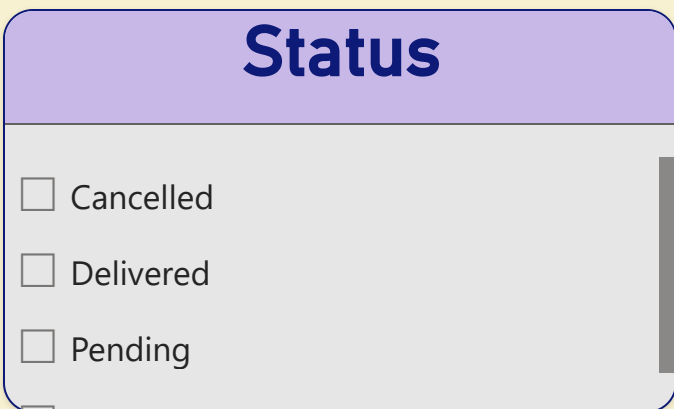
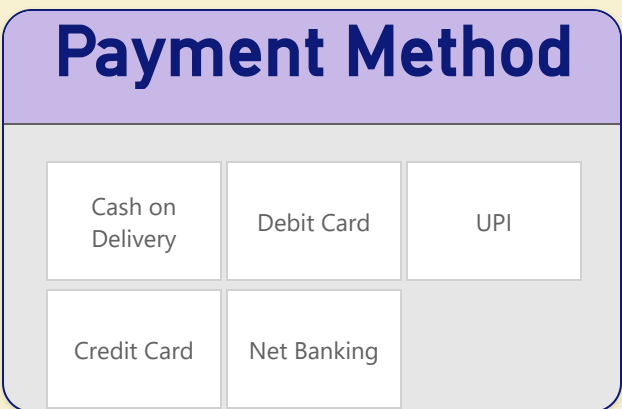
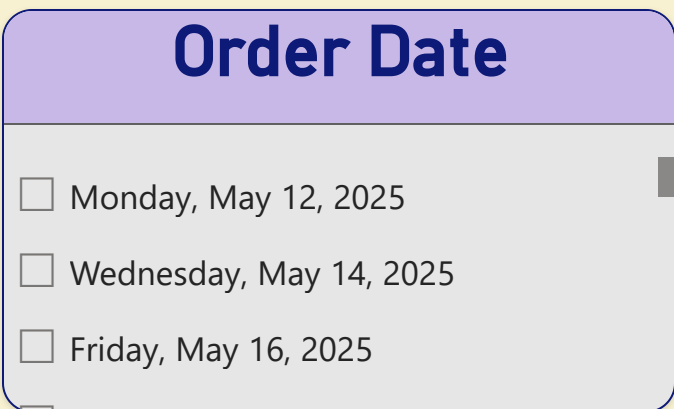
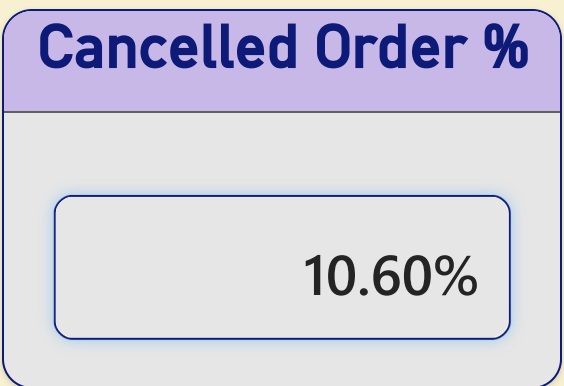
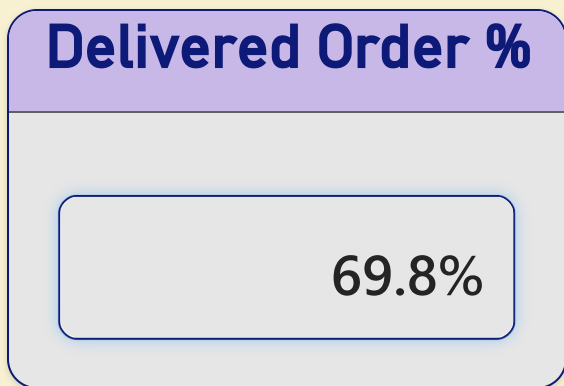


Products Requiring Business Attention				
Product_Name	Category	Total Quantity Sold	Total Revenue	Average Product Price
Classic Eius	Books	123	\$1,554,641.28	12,639.36
Classic Eos	Electronics	93	\$1,377,823.83	14,815.31
Classic Incidunt	Sports	76	\$3,273,003.08	43,065.83
Classic Ipsa	Electronics	81	\$1,019,149.29	12,582.09
Classic Numquam	Clothing	84	\$259,028.28	3,083.67
Classic Repellendus	Sports	115	\$4,386,642.8	38,144.72
Classic Tempora	Electronics	115	\$2,967,456.55	25,803.97
Classic Unde	Books	82	\$3,689,981.96	44,999.78
Eco A	Sports	102	\$5,080,195.68	49,805.84
Eco Aut	Home & Kitchen	97	\$2,062,682.69	21,264.77
Eco Consectetur	Books	88	\$851,964.08	9,681.41
Total		4549	\$116,154,185.82	25,497.10

Product & Category Performance – Business Insights

- **Electronics** is the highest revenue-generating category, followed by **Sports** and **Books**.
- Product quantity sold is distributed across several products, with **Classic Eco** and other Classic-series products among the higher-volume products shown.
- **Top products contribute a significant share of category revenue**, indicating the importance of best-performing products.
- Several products appear in the **Low-Selling Products** visual, indicating that their sales performance needs attention.
- The **Low Stock Products** visual highlights products with limited inventory, so stock levels should be monitored to avoid potential stock-outs.

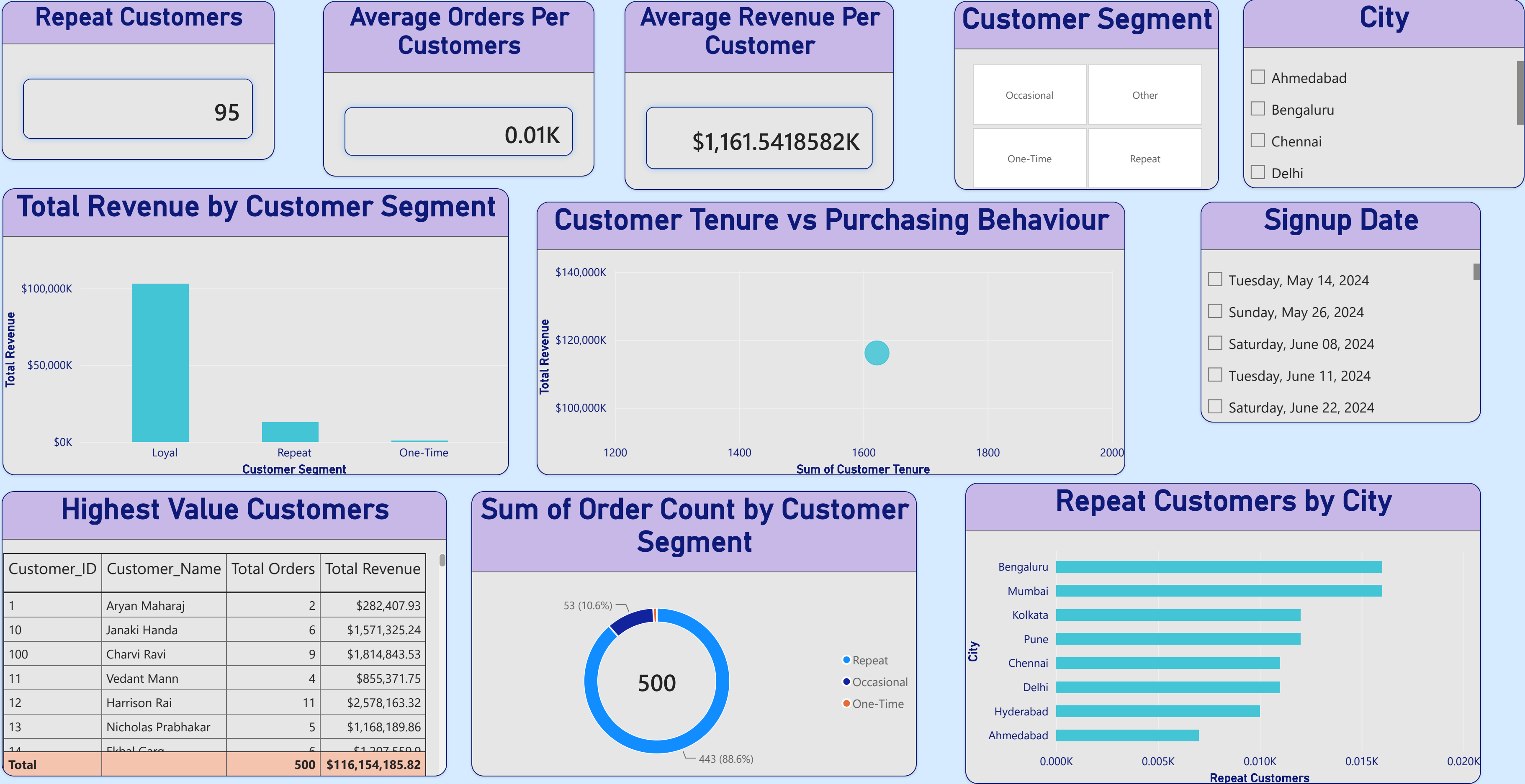
Order & Payment Behaviour



Order & Payment Behaviour – Business Insights

- **69.8% of orders are delivered**, indicating successful order fulfilment.
- **10.60% cancellation rate** indicates a need to investigate cancelled orders.
- **Net Banking generates the highest revenue** among payment methods.
- **Cash on Delivery has the highest cancelled orders** and requires attention. Cancellation rates vary across cities, highlighting potential operational issues.

Customer Value & Retention



Total Revenue by Customer Segment

Customer Segment	Total Revenue (\$K)
Loyal	~100,000
Repeat	~10,000
One-Time	~1,000

Customer Tenure vs Purchasing Behaviour

Sum of Customer Tenure	Total Revenue (\$K)
~1650	~116,154.18582

Signup Date

☐ Tuesday, May 14, 2024

☐ Sunday, May 26, 2024

☐ Saturday, June 08, 2024

☐ Tuesday, June 11, 2024

☐ Saturday, June 22, 2024

Highest Value Customers

Customer_ID	Customer_Name	Total Orders	Total Revenue
1	Aryan Maharaj	2	\$282,407.93
10	Janaki Handa	6	\$1,571,325.24
100	Charvi Ravi	9	\$1,814,843.53
11	Vedant Mann	4	\$855,371.75
12	Harrison Rai	11	\$2,578,163.32
13	Nicholas Prabhakar	5	\$1,168,189.86
14	Ekbal Garg	6	\$1,207,550.00
Total		500	\$116,154,185.82

Sum of Order Count by Customer Segment

Customer Segment	Order Count	Percentage
Repeat	443	88.6%
Occasional	53	10.6%
One-Time	4	0.8%

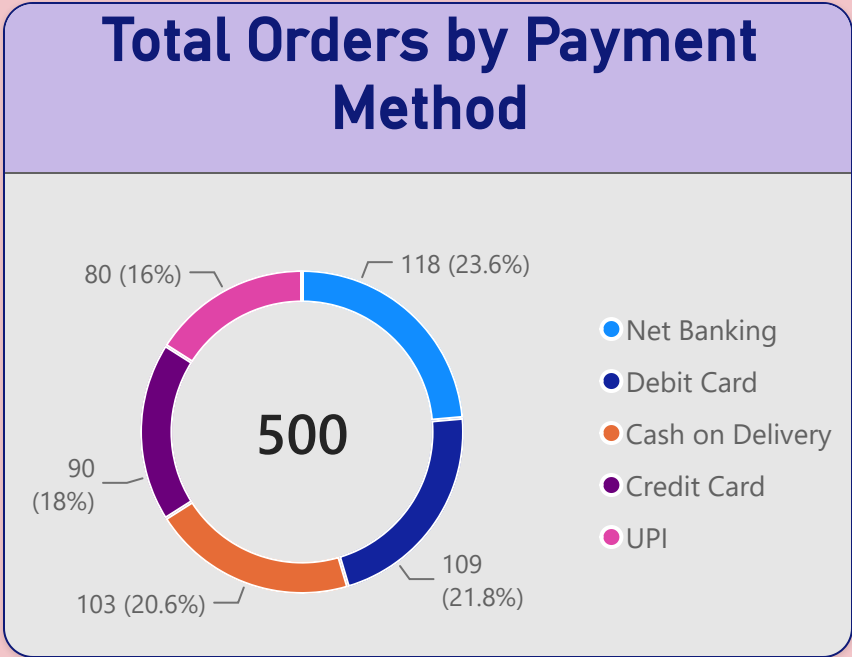
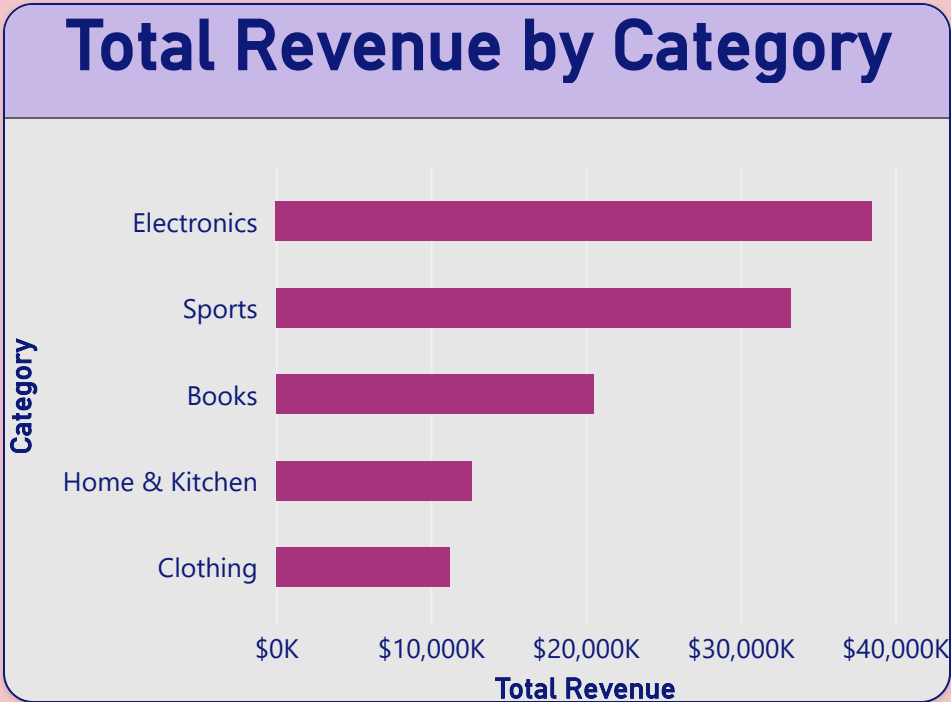
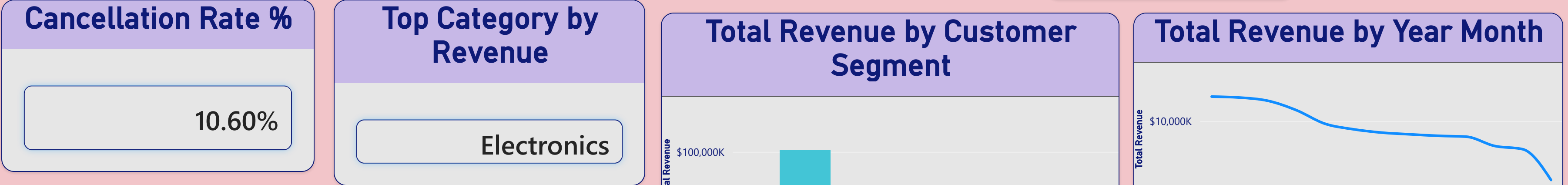
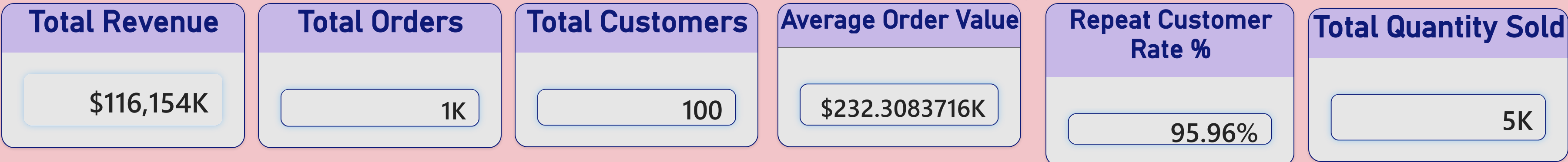
Repeat Customers by City

City	Repeat Customers (K)
Bengaluru	~0.016
Mumbai	~0.016
Kolkata	~0.012
Pune	~0.012
Chennai	~0.011
Delhi	~0.011
Hyderabad	~0.010
Ahmedabad	~0.007

Customer Value & Retention – Business Insights

- **95 repeat customers** indicate strong customer retention.
- **Loyal customers generate the highest revenue** among the segments.
- Average revenue per customer is approximately **\$1.16M**.
- High-value customers contribute significantly to overall revenue and should be prioritized for retention.
- Repeat-customer concentration varies across cities, creating opportunities for targeted retention strategies.
- Customer tenure and purchasing behaviour can help identify high-value long-term customers.

Executive Summary



Executive Insights:

- Top category: Electronics generates the highest revenue.
- Customer Segment: Repeat Contributes the highest revenue.
- Payment Method: Net Banking is the dominant payment method.
- Cancellation: Cancellation rate is 10.60%, indicating Net Banking.
- Product Requiring Attention: Clothing shows relatively low stock and requires attention through promotion planning.
- Opportunity: Mumbai generates the highest revenue, indicating strong customer demand and a potential opportunity for further business growth.

Business Recommendations

1. Focus marketing efforts on high-performing categories to maximize revenue growth.
2. Strengthen customer retention strategies by offering personalized offers and loyalty benefits to repeat and high-value customers.
3. Investigate high cancellation rates and identify the major cities, payment methods, or order patterns contributing to cancellations.
4. Improve inventory planning for high-demand or low-stock products to avoid stock-related sales opportunities being missed.
5. Expand business efforts in high-potential cities by increasing promotions and customer acquisition activities where revenue and orders are strong.
6. Promote high-performing products while reviewing low-performing products for pricing, visibility, or promotional improvements.
7. Optimize payment options based on customer preferences and encourage convenient payment methods to improve order completion.